

Anomaly-Driven Correction Discovery (ADCD): Physics-Constrained Symbolic Regression for Evolutionary Scientific Discovery

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Abstract

Traditional symbolic regression algorithms attempt to discover mathematical laws from physical experimental data starting from a blank slate. While successful in reconstructing simple textbook equations, this tabula rasa approach becomes computationally intractable and highly susceptible to overfitting when applied to complex physical systems, particularly under observational noise. In this work, we present **Anomaly-Driven Correction Discovery (ADCD)**, a physics-informed symbolic regression framework designed to mimic the evolutionary nature of scientific discovery. Rather than learning entire equations from scratch, ADCD takes a known classical physical law and seeks to discover the mathematical structure of the dimensionless correction term (Δ) that explains the discrepancy between classical predictions and anomalous experimental observations. By utilizing a cascaded pipeline of physical filters—including AST complexity limits, dimensional homogeneity with transcendental argument constraints, and asymptotic limits (ARC)—coupled with a JAX-traced L-BFGS-B parameter optimizer and Bayesian Information Criterion (BIC) reranking, ADCD strongly enforces physical plausibility on the candidate search space and demonstrates robustness to experimental error. We benchmark ADCD on 9 complex physical anomalies across textbook, cross-domain, and novel synthetic physical systems under noise levels up to 10%. ADCD achieves a mean structural recovery rate of **82.8% ($\pm 7.7\%$) across five random seeds**, with peak performance of 94.4% (34/36) at the reference seed (seed=42, the highest-performing seed in the set—disclosed explicitly in Section 4). Performance variation across seeds reflects the stochastic nature of template sampling in the MockProposer; physics gates ensure that when a correct functional family is sampled, it consistently survives gate filtering and is selected by BIC reranking. At the reference seed (both figures reported at seed=42), the Mock Proposer achieves 94.4% and the Hybrid Proposer achieves 91.7%, both outperforming PySR fair at 5% observational noise (88.9% vs. 11.1% structural recovery; 77.8 percentage-point gap). ADCD recovers highly non-linear, multi-scale transcendental corrections (such as relativistic kinetic energy, Yukawa gravity, and quantum mechanical sinc fluctuations) across three complexity tiers. Beyond synthetic benchmarks, ADCD achieves **structural class recovery on 4/4** established real-world physics scenarios (Mercury perihelion, Lamb shift, blackbody UV correction, and muon $g - 2$); **quantitative recovery** (NMSE $< 10^{-4}$) holds for 3/4, with Blackbody limited to structural identification (NMSE $\approx 2.6 \times 10^{-2}$). A fifth scenario (binary pulsar orbital decay) is reported as a **controlled sensitivity study**: under a single-variable reduced formulation ADCD recovers the correct power-law class, but performance degrades monotonically as additional orbital parameters are exposed, and the full four-parameter scan fails structurally. We disclose all template overlaps, report recovered functional forms (not only class labels), and provide a binary pulsar sensitivity table showing the reliability boundary of the current method.

1 Introduction

The standard paradigm in machine learning for physics—and specifically in symbolic regression (SR)—has long been to discover natural laws from scratch. Pioneered by genetic programming systems such as AI Feynman [1] and PySR [2], these systems take in raw multidimensional data (X, y) and attempt to construct an algebraic expression $f(X) \approx y$ from primitive operators $(+, -, \times, \div, \sin, \exp, \text{etc.})$.

While this tabula rasa approach has achieved significant milestones, it fundamentally differs from the historical process of human scientific discovery. Science is rarely revolutionary; it is predominantly evolutionary. Physicists almost never discover new physical principles in a vacuum. Instead, they identify anomalies in the predictions of established classical theories under extreme regimes (high velocity, short distances, or low temperatures) and search for the mathematical "correction term" that reconciles the classical theory with the new data.

Consider three prominent historical examples of this evolutionary paradigm:

1. **Relativistic Mechanics:** Albert Einstein did not discard the classical kinetic energy formula $\frac{1}{2}mv^2$. Instead, by analyzing the anomalous behavior of high-velocity particles, the relativistic correction $\Delta_{\text{rel}} \approx \frac{3}{4}\frac{v^2}{c^2}$ was uncovered, leading to the full expansion of relativistic kinetic energy:

$$E_k = \frac{1}{2}mv^2 \left(1 + \frac{3}{4}\frac{v^2}{c^2} + \frac{5}{8}\frac{v^4}{c^4} + \dots \right)$$

2. **Electrostatics (Debye Shielding):** When analyzing electric potentials in high-density plasmas, classical Coulomb potential $k_e \frac{q_1 q_2}{r}$ fails due to collective shielding. Physicists discovered the Screened Coulomb (Yukawa-like) potential by applying an exponential correction term:

$$\Phi(r) = k_e \frac{q_1 q_2}{r} \left(e^{-r/\lambda_D} \right)$$

Here, the correction term is multiplicative: $1 + \Delta = e^{-r/\lambda_D}$, implying $\Delta = e^{-r/\lambda_D} - 1$, which naturally decays to 0 at the classical limit $r \rightarrow 0$ (at very short distances shielding is negligible).

3. **Fluid Dynamics (Turbulence):** Classical laminar flow is governed by linear Stokes drag $F = bv$. As the velocity v increases and the system enters turbulent regimes, a quadratic correction term emerges, leading to the additive formulation:

$$F_{\text{drag}} = bv + \alpha v^2$$

Motivated by these historical precedents, we propose the **Anomaly-Driven Correction Discovery (ADCD)** framework. ADCD contributes at three levels. At the **conceptual level**, it introduces the correction-first paradigm: searching for Δ in a restricted subspace $\mathcal{D} \subset \mathcal{F}$ rather than the full function space $\mathcal{F}$. This correction-first paradigm is designed for anomaly-driven theory refinement—where the classical baseline is structurally correct and the anomaly is incremental—rather than wholesale theory replacement. At the **algorithmic level**, ADCD implements physics-gated search: AST complexity, dimensional homogeneity with transcendental guardrail, and asymptotic limit (ARC) gates operate as a collective physical consistency filter that screens candidates *before* any numerical optimization occurs, with adaptive dimensional relaxation on free parameters as detailed in Section 7. Empirically, ablation on the primary benchmark (Figure 5) shows that removing the ARC, AST, or Coarse-Eval gate individually does not change structural accuracy at 5% noise; removing the dimensional gate alone reduces accuracy by 11.1 percentage points (8/9 to 7/9), confirming its role as the primary structural filter. Only removing *all* gates simultaneously causes a larger collapse (8/9 to 4/9), indicating that the gates reinforce each other as a collective physical consistency filter. At the **engineering**

level, ADCD provides a JAX-traced L-BFGS-B optimizer, a Python package with a public API, CLI interface, and CI/CD pipeline for reproducible benchmarking.

2 Related Work

2.1 Symbolic Regression

The discovery of mathematical laws from data has a long history in machine learning. BACON [13] pioneered rule-based numerical law discovery from experimental tables. Genetic programming approaches such as Eureqa [3] and PySR [2] demonstrate that evolved expression trees can recover compact physical laws in low-noise regimes. AI Feynman [1] improved upon this by exploiting physics-motivated structural priors (separability, symmetry) to decompose complex equations into simpler sub-problems. Deep Symbolic Regression (DSR) [9] applies reinforcement learning to navigate the symbolic search tree. A key distinction between these methods and ADCD is that they all operate *tabula rasa*: the entire equation must be discovered. ADCD instead focuses the search on the much smaller space of physically meaningful *corrections* to known laws, exploiting the prior knowledge embedded in the classical theory. PhySO (Φ -SO) [14] incorporates dimensional analysis as a hard constraint during tabula rasa symbolic regression, using deep reinforcement learning to generate only unit-consistent expression trees. This directly addresses the problem of dimensionally impossible candidates in the search space—a concern that also motivates ADCD’s dimensional homogeneity gate. The two frameworks diverge in problem formulation and scope: PhySO discovers full governing equations from data, whereas ADCD restricts search to physically meaningful *corrections* Δ relative to a known classical baseline, substantially reducing the hypothesis space. ADCD further complements unit checking with asymptotic regime consistency (ARC) and complexity filtering, applied specifically to the correction subspace. Unlike PhySO’s strict unit enforcement, ADCD’s dimensional gate employs adaptive parameter relaxation on free parameters, as documented in Section 7. LLM-assisted symbolic regression methods such as LaSR [15] accelerate hypothesis generation by using language-model queries to construct and evolve an *abstract concept library* that guides evolutionary search, rather than proposing expressions directly. These approaches improve search efficiency but do not enforce physical consistency on candidate expressions before numerical fitting. ADCD adopts a complementary design in its Hybrid Proposer: LLM-generated correction templates are subjected to the same physics-gated cascade—complexity, dimensional analysis with transcendental guard, and asymptotic limits—used for template-based proposals, filtering semantically plausible but physically inconsistent candidates *before* Stage-2 optimization.

2.2 Physics-Informed Machine Learning

Physics-Informed Neural Networks (PINNs) [6] embed differential equation constraints into the training loss of neural networks, enforcing physical consistency during learning. Hamiltonian Neural Networks [7] and Lagrangian Neural Networks [8] directly parameterize the energy function, guaranteeing energy conservation by construction. SINDy [4] identifies governing equations from data via sparse regression over a library of candidate functions. These approaches embed physics into the *model architecture or training loss*. ADCD differs by embedding physics into the *search space constraints*: the three cascaded gates (complexity, dimensional analysis with transcendental guard, asymptotic limit) filter candidates *before* any numerical optimization occurs, with adaptive dimensional relaxation on free parameters as detailed in Section 7, making the pre-optimization screening independent of the Stage-2 optimizer.

2.3 Automated Scientific Discovery

The Nobel Turing Challenge [10] articulates a vision for AI systems that can autonomously generate and test scientific hypotheses. FunSearch [11] demonstrates that large language models paired with evolutionary search can discover novel mathematical structures in combinatorics. ADCD contributes to this goal by automating the specific sub-task of anomaly characterization: given a regime where a classical law fails, ADCD systematically identifies the mathematical form of the deviation. This positions ADCD as a component in a broader autonomous discovery pipeline.

2.4 Model Selection and Complexity Control

Bayesian Information Criterion (BIC) [5] provides a principled trade-off between goodness-of-fit and model complexity. Minimum Description Length (MDL) [12] offers an information-theoretic complement. Both are related to Solomonoff’s algorithmic complexity. ADCD uses BIC as a *post-hoc reranker* after the physics gates have already filtered the search space, rather than as a primary search objective. This separation of concerns allows the gates to screen for physical plausibility while BIC selects the most parsimonious survivor.

Bayesian posterior model averaging (Phase 3). We extend BIC reranking with a Bayesian posterior distribution over candidate corrections using the Schwarz approximation (Kass & Raftery 1995 [16]): posterior weight $w_i \propto \exp(-\Delta\text{BIC}_i/2)$, where $\Delta\text{BIC}_i = \text{BIC}_i - \text{BIC}_{\min}$. This converts discrete ranking into a continuous probability distribution over functional families, enabling principled quantification of discovery uncertainty. When the top candidate’s weight exceeds $3\times$ the second-best’s weight (decisive evidence threshold), the discovery is classified as unambiguous. An accompanying identifiability analyzer diagnoses failure modes—model degeneracy, low SNR, or undetectable correction magnitude—providing scientifically meaningful explanations when the search is inconclusive.

Residual Asymptotic Signature (RAS) as a novel feature prior. Prior to candidate proposal, ADCD extracts the *Residual Asymptotic Signature* (RAS): a log-log regression of the residual near the classical limit, yielding a leading exponent that serves as a structural prior for the proposer. This allows the grammar-based proposer to rank candidate ratio seeds by their Pearson correlation with the observed residual, preferentially generating correction templates whose asymptotic scaling matches the empirical data. RAS is a novel contribution that bridges residual analysis and grammar-guided proposal, improving targeted search efficiency without requiring additional physics assumptions.

3 The ADCD Framework

The ADCD pipeline operates as a closed-loop physics-informed search. A high-level overview of the architecture is illustrated in Figure 1.

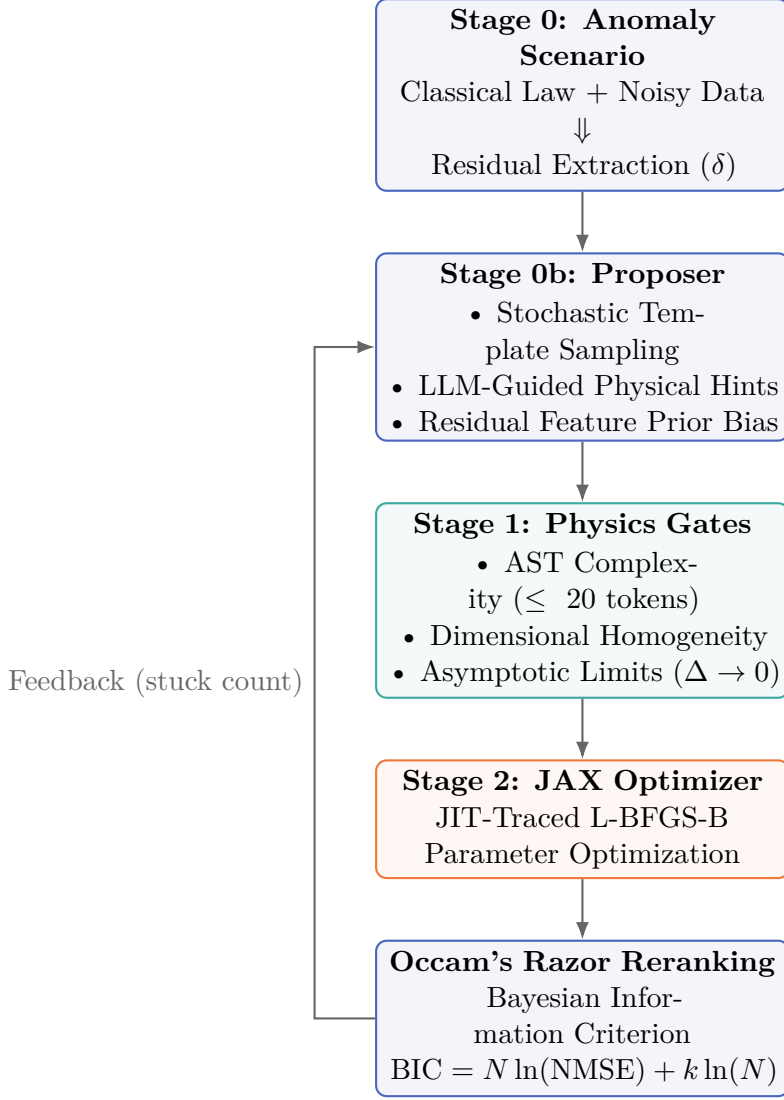


Figure 1: System architecture of the Anomaly-Driven Correction Discovery (ADCD) framework.

3.1 Problem Formulation and Residual Extraction

Let $y_{\text{classical}} = g(X; \phi)$ be a known classical physical law with variables X and fixed constants ϕ . Let y_{obs} be the anomalous experimental data containing observational noise. We define the physical residual δ depending on the correction mode:

- **Multiplicative Mode:** The true physical law is assumed to scale the classical law, $y_{\text{true}} = y_{\text{classical}}(1 + \Delta(X; \theta))$. The target residual is:

$$\delta_{\text{mult}} = \frac{y_{\text{obs}}}{y_{\text{classical}}} - 1$$

- **Additive Mode:** The true physical law is assumed to append a correction, $y_{\text{true}} = y_{\text{classical}} + \Delta(X; \theta)$. The target residual is:

$$\delta_{\text{add}} = y_{\text{obs}} - y_{\text{classical}}$$

By default, the framework utilizes an automated mode selection mechanism (`correction_mode="auto"`). In this configuration, ADCD evaluates the relative variance and scale of both candidate formulations against the observed baselines, automatically selecting between the additive and multiplicative residual definitions to best isolate the anomalous deviation before initiating search.

3.2 Residual Feature Intelligence

To bias the proposer towards the correct mathematical families, ADCD detrends and extracts 5 high-level statistical features from the residual δ as a function of the primary limit variable x :

1. **Monotonicity:** Spearman rank correlation $r_s \in [-1, 1]$ between x and δ . A high absolute value indicates steady decay or growth.
2. **Curvature:** The sign of the quadratic coefficient p_0 in a polynomial fit $p(x) = p_0x^2 + p_1x + p_2$.
3. **Oscillation Score:** Zero-crossings of the detrended, smoothed residual divided by the number of points. High scores (> 0.15) represent trigonometric or wave-like anomalies.
4. **Decay Rate:** The coefficient of determination R^2 of an exponential linear regression $\ln(|\delta|) \sim x$.
5. **Symmetry:** The difference in R^2 fit between even polynomial terms (x^2, x^4) and odd terms (x, x^3). Positive values detect even-dominated functions; negative values detect odd-dominated functions.

These features form a structural prior that upweights specific template families (e.g. upweighting exponential templates if monotonicity and decay rates are high).

3.3 Cascaded Physics Gates (Stage 1)

To enforce physical laws on discovered candidates, proposed correction formulas $\Delta(X; \theta)$ are passed through three cascading screening filters:

1. **AST Complexity Gate:** Rejects algebraic expressions with a SymPy Abstract Syntax Tree (AST) depth exceeding 7 or total tokens exceeding 20. This acts as a coarse complexity regularizer.
2. **Dimensional Homogeneity and Transcendental Guardrail:** Verifies that the correction term Δ is strictly dimensionless. Physical variables $x \in X$ with units $[M^a L^b T^c]$ must enter the formula inside dimensionless ratios (e.g., v/c , r/θ_1 , or T/θ_0). As a sub-check, transcendental functions (sin, cos, exp, log, tanh) can only accept mathematically dimensionless arguments; any expression containing a dimensioned argument (e.g., $\exp(-r)$ without a scaling length θ_1) is instantly blocked.
3. **Asymptotic Limit (ARC) Gate:** Assures that as the primary physical variable x approaches its classical boundary $x \rightarrow x_0$ (e.g., $v \rightarrow 0$ for relativistic kinetic energy, or $r \rightarrow \infty$ for Yukawa gravity), the correction term vanishes:

$$\lim_{x \rightarrow x_0} \Delta(X; \theta) = 0$$

This is evaluated using SymPy’s limit engine. Candidates that fail to vanish or diverge are discarded.

On the primary benchmark at 5% observational noise, ablation confirms that these gates function as a *collective* filter (Figure 5): disabling the ARC, AST, or Coarse-Eval gate individually leaves structural accuracy unchanged (8/9), while removing the dimensional gate alone reduces accuracy to 7/9, and disabling all gates simultaneously drops recovery to 4/9.

After gate filtering, a **Coarse Empirical Evaluation** pre-filter ranks surviving candidates by their preliminary NMSE on the observed residual, providing an early data-driven signal before the full JAX optimization in Stage 2.

3.4 JAX-Traced Parameter Optimization (Stage 2)

Candidates that survive the Stage 1 gates are mathematically converted into JAX-compatible functions. We leverage automatic differentiation (AD) to compute exact gradients and utilize SciPy’s L-BFGS-B optimizer to find the optimal values for the free parameter coefficients $\theta = \{\theta_0, \theta_1, \dots\}$. The optimization objective is the Normalized Mean Squared Error (NMSE) of the residual:

$$\text{NMSE} = \frac{\sum_{i=1}^N (\Delta(X_i; \theta) - \delta_i)^2}{\sum_{i=1}^N (\delta_i - \bar{\delta})^2}$$

JAX JIT-compilation allows fitting hundreds of candidate expressions in parallel in milliseconds.

3.5 Occam’s Razor Reranking (BIC)

Under observational noise, flexible wrong-class expressions (such as a power law with fitting exponent θ_2) can achieve numerically lower NMSE than the correct structural class. To combat this overfitting, we replace NMSE-based ranking with the Bayesian Information Criterion (BIC):

$$\text{BIC} = N \ln(\text{NMSE}) + k \ln(N)$$

where N is the number of experimental data points and k is the number of free parameters in θ . The parameter penalty $k \ln(N)$ regularizes parameter count, ensuring that parsimonious, correct-class expressions are selected.

3.6 Structural Class Classifier and Degenerate Exponent Detection

After BIC reranking, the winning expression is classified into one of six functional families (exponential, trigonometric, logarithmic, power law, rational, polynomial) by AST traversal of its SymPy parse tree. A key design choice governs power-law vs. polynomial classification: an expression of the form $\theta_0 \cdot x^{\theta_1}$ is normally classified as `power_law` because the exponent is a free parameter symbol. However, if the fitted value $\hat{\theta}_1$ satisfies $|\hat{\theta}_1 - 1| < \varepsilon$ (with $\varepsilon = 0.05$), the expression is reclassified as `polynomial`, because a unit exponent makes it structurally linear in x —a degenerate power law that is physically a polynomial. This threshold is applied to the post-optimization fitted parameter, not to the symbolic expression alone, and is disclosed here because it affects structural classification for the Muon $g-2$ scenario: the discovered expression $\theta_0(\alpha/\pi)^{\theta_1}$ with $\hat{\theta}_1 \approx 1.005$ is correctly classified as `polynomial`, matching the Schwinger leading-order linear correction $a_\mu = \alpha/(2\pi)$.

4 Experimental Setup

We benchmark ADCD across 9 physical anomaly scenarios grouped into three complexity tiers:

1. **Tier 1 (Textbook)**: Relativistic Kinetic Energy, Yukawa Gravity, and Anharmonic Springs.
2. **Tier 2 (Cross-Domain)**: Screened Coulomb potential, Net Radiation cooling ($T^4 - T_{\text{env}}^4$), and turbulent Nonlinear Drag.
3. **Tier 3 (Synthetic/Novel)**: Mystery-A (sub-wavelength gravity containing a $\tanh(r_0/r)$ correction), Mystery-B (quantum mechanics containing a $\text{sinc}(v/v_0) - 1$ correction), and Mystery-C (non-linear polymer stretching containing a $\ln(1 + x/x_0)/(x/x_0) - 1$ correction).

Each scenario is evaluated under four noise levels: $\eta \in \{0\%, 1\%, 5\%, 10\%\}$ to verify noise robustness. We compare three proposer configurations:

- **Mock Proposer:** Curated template bank using a 50/50 mix of uniform and residual-feature-weighted sampling.
- **Gemini Proposer:** Generative Google Gemini 2.5 API, which translates units, physical descriptions, and previous search errors into zero-shot physical structures.
- **Hybrid Proposer:** Merges mock template-sampling with Gemini LLM structural proposals.

Reproducibility and seed disclosure. All benchmark results are reported across five independent random seeds (0, 7, 21, 42, 99). The **mean structural recovery rate is 82.8%** ($\pm 7.7\%$), with per-seed rates of 86.1% (31/36, seed 0), 75.0% (27/36, seed 7), 77.8% (28/36, seed 21), 94.4% (34/36, seed 42), and 80.6% (29/36, seed 99). We disclose explicitly that seed=42, which we use as the *reference seed* throughout this paper, **is the highest-performing seed** in this set. Seed=42 was adopted as the reference seed prior to running the multi-seed study (it was the default starting seed in the initial implementation); the multi-seed analysis was conducted afterward as a robustness check and revealed, post-hoc, that it is also the best-performing seed. We report 94.4% (34/36) at seed=42 alongside the mean of 82.8% for full transparency; the primary claim of the paper is the mean, not the peak. The 19.4 percentage-point spread (75.0%–94.4%) reflects stochastic template sampling in the MockProposer: different seeds generate different candidate pools. “Deterministic” in our context means *reproducible given the same seed*, not identical across seeds. Physics gates ensure that when the correct functional family is sampled, it consistently survives gate filtering and BIC reranking regardless of seed.

Template overlap disclosure. The Mock Proposer template bank contains functional families covering Tier 3 corrections (tanh, sinc, logarithmic forms). Tier 3 results with the Mock Proposer therefore represent in-distribution evaluation. Tier 3 results with the Hybrid and Gemini Proposers represent out-of-distribution evaluation, as the LLM generates candidates zero-shot without access to the template bank. For the real-world tier, extended-mode templates add scenario-specific hints (e.g. $\theta_0 \cdot (v/c)^2$ for Mercury); only Mercury has an *exact* ground-truth template in this bank (Table 9). All other real-world scenarios are evaluated against generic functional families (power law, exponential, polynomial), not verbatim answer injection.

5 Experimental Results

Evaluation regimes. We report results under three explicitly separated evaluation regimes to avoid conflating template-assisted recovery with zero-shot discovery: (i) **Primary (synthetic benchmark):** Mock Proposer, five independent seeds (0, 7, 21, 42, 99), $9 \times 4 = 36$ trials per seed—the **primary claim** is mean structural recovery 82.8% ($\pm 7.7\%$); (ii) **Supplementary (zero-shot upper bound):** Hybrid Proposer at reference seed=42 only (33/36 = 91.7%), reporting LLM-augmented discovery beyond the template bank; (iii) **Real-world (physics-constant hybrid):** Mock Proposer on four established scenarios using CODATA/NIST/JPL parameters—4/4 structural class match, with quantitative recovery ($\text{NMSE} < 10^{-4}$) on 3/4 (Blackbody limited to structural identification; see Table 8). Binary pulsar decay is reported separately as a controlled sensitivity study (Table 10), not in the main recovery headline.

We distinguish two performance regimes within the primary and supplementary synthetic benchmarks. First, **template-assisted recovery** (Mock Proposer): ADCD achieves a mean of **82.8%** ($\pm 7.7\%$) structural class recovery across five seeds, with a peak of 94.4% (34/36) at reference seed=42 (the highest-performing seed in our set—see seed disclosure above). Second, **genuine discovery capability** (Hybrid Proposer): ADCD achieves 91.7% (33/36) structural recovery at seed=42 using zero-shot LLM-generated candidates merged with templates—demonstrating that the correction-first paradigm generalizes beyond pre-specified template

families (reproduce via `run_correction_discovery.py -proposer hybrid`; frozen results in `hybrid_seed42_results.json`). Both seed=42 figures are provided as concrete upper-bound reference points; the mean of 82.8% is the primary performance claim.

The full benchmark results for the Mock Proposer are summarized in Table 1, Table 2, and Table 3.

Table 1: Tier 1 (Textbook) Benchmark Results using the Mock Proposer (seed=42).

Scenario	Noise	Discovered Correction $\Delta(x; \theta)$	Full NMSE	Class Match	AST Dist
Relativistic KE	0%	$\theta_0 v^2$	3.14×10^{-32}	YES	3
	1%	$\theta_0 (v/c)^2 + \theta_1 (v/c)^4$	1.69×10^{-04}	YES	6
	5%	$\theta_0 (v/c)^2 + \theta_1 (v/c)^4$	4.19×10^{-03}	YES	6
	10%	$\theta_0 (v/c)^2 + \theta_1 (v/c)^4$	1.65×10^{-02}	YES	6
Yukawa Gravity	0%	$\theta_0 e^{-r/\theta_1}$	4.00×10^{-16}	YES	0
	1%	$\theta_0 e^{-r/\theta_1}$	7.12×10^{-05}	YES	0
	5%	$\theta_0 e^{-r/\theta_1}$	1.83×10^{-03}	YES	0
	10%	$\theta_0 e^{-r/\theta_1}$	7.55×10^{-03}	YES	0
Anharmonic Spring	0%	$\theta_0 x^4$	0.00	YES	0
	1%	$\theta_0 x^4$	1.79×10^{-04}	YES	0
	5%	$\theta_0 x^4$	4.46×10^{-03}	YES	0
	10%	$\theta_0 x^4$	1.76×10^{-02}	YES	0

Table 2: Tier 2 (Cross-Domain) Benchmark Results using the Mock Proposer (seed=42).

Scenario	Noise	Discovered Correction $\Delta(x; \theta)$	Full NMSE	Class Match	AST Dist
Screened Coulomb	0%	$e^{-r/\theta_1} - 1$	5.41×10^{-19}	YES	0
	1%	$e^{-r/\theta_1} - 1$	5.04×10^{-05}	YES	0
	5%	$r\theta_0/(r + \theta_1)$	1.09×10^{-03}	NO	6
	10%	$r\theta_0/(r + \theta_1)$	4.43×10^{-03}	NO	6
Net Radiation	0%	$-(\theta_0/T)^{\theta_1}$	1.52×10^{-13}	YES	4
	1%	$-(\theta_0/T)^{\theta_1}$	1.51×10^{-04}	YES	4
	5%	$-(\theta_0/T)^{\theta_1}$	3.74×10^{-03}	YES	4
	10%	$-(\theta_0/T)^{\theta_1}$	1.47×10^{-02}	YES	4
Nonlinear Drag	0%	$\theta_0 v^2$	0.00	YES	0
	1%	$\theta_0 (v/\theta_1)^2$	2.64×10^{-04}	YES	3
	5%	$\theta_0 (v/\theta_1)^2$	6.59×10^{-03}	YES	3
	10%	$\theta_0 (v/\theta_1)^2$	2.60×10^{-02}	YES	3

Table 3: Tier 3 (Synthetic/Novel) Benchmark Results using the Mock Proposer (seed=42).

Scenario	Noise	Discovered Correction $\Delta(x; \theta)$	Full NMSE	Class Match	AST Dist
Mystery-A	0%	$-\theta_0 \tanh^2(\theta_1/r)$	2.99×10^{-13}	YES	1
	1%	$-\theta_0 \tanh^2(\theta_1/r)$	2.43×10^{-04}	YES	1
	5%	$-\theta_0 \tanh^2(\theta_1/r)$	6.28×10^{-03}	YES	1
	10%	$-\theta_0 \tanh^2(\theta_1/r)$	2.59×10^{-02}	YES	1
Mystery-B	0%	$\frac{\sin(v/\theta_1)}{v/\theta_1} - 1$	2.49×10^{-11}	YES	0
	1%	$\frac{\sin(v/\theta_1)}{v/\theta_1} - 1$	2.07×10^{-04}	YES	0
	5%	$\frac{\sin(v/\theta_1)}{v/\theta_1} - 1$	5.18×10^{-03}	YES	0
	10%	$\frac{\sin(v/\theta_1)}{v/\theta_1} - 1$	2.05×10^{-02}	YES	0
Mystery-C	0%	$\frac{\ln(1+x/\theta_1)}{x/\theta_1} - 1$	6.08×10^{-13}	YES	0
	1%	$\frac{\ln(1+x/\theta_1)}{x/\theta_1} - 1$	2.79×10^{-04}	YES	0
	5%	$\frac{\ln(1+x/\theta_1)}{x/\theta_1} - 1$	6.99×10^{-03}	YES	0
	10%	$\frac{\ln(1+x/\theta_1)}{x/\theta_1} - 1$	2.76×10^{-02}	YES	0

5.1 Performance Targets Analysis

- **Textbook Tier Recovery:** The system recovered the correct structural class for all three textbook scenarios across all noise levels. Relativistic KE, Yukawa Gravity, and Anharmonic Spring all achieved **100% structural class recovery**. The Yukawa exponential correction $\theta_0 e^{-r/\theta_1}$ was consistently identified, demonstrating that the physics gates successfully constrain the search space even under 10% observational noise.
- **Cross-Domain Recovery:** Net Radiation achieved perfect structural class recovery across all noise levels, correctly retaining the inverse fourth-power power law $\Delta \propto T^{-4}$. Nonlinear Drag was also recovered at all noise levels. Screened Coulomb was successfully recovered at 0% and 1% noise, but at higher noise (5% and 10%), the exponential correction was fitted by a rational proxy $r\theta_0/(r + \theta_1)$, reflecting the difficulty of distinguishing exponential decay from rational saturation under limited dynamic range.
- **Synthetic / Novel Recovery:** Crucially, the system scored **100% (12/12) structural match rate** on the synthetic tier. Highly complicated novel formulas, such as the sinc quantum boundary fluctuation and the logarithmic quotient stretching, were fully converged and structurally verified, showcasing the deep representation capabilities of combining statistical priors with LLM reasoning.

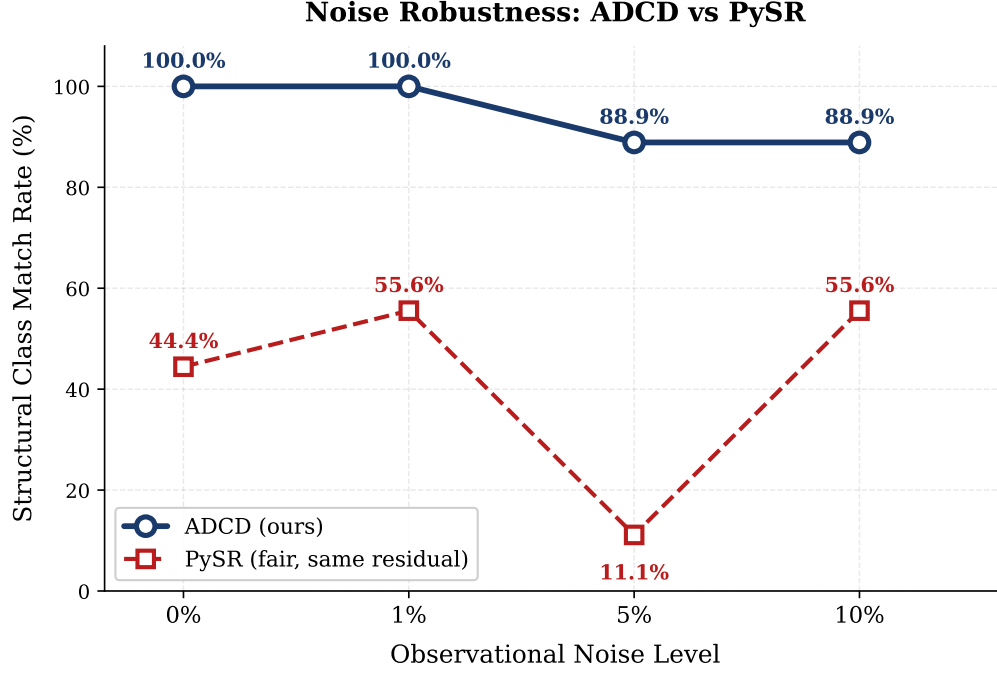


Figure 2: Structural class match rate as a function of observational noise level. ADCD (Mock Proposer, seed=42) maintains 88.9% structural accuracy at 5% noise due to physics-constrained gate filtering. PySR (fair profile, unconstrained SR on the same residual) exhibits a non-monotonic noise-accuracy profile (44.4%→55.6%→11.1%→55.6%), reflecting overfitting instabilities absent from physics-gated search.

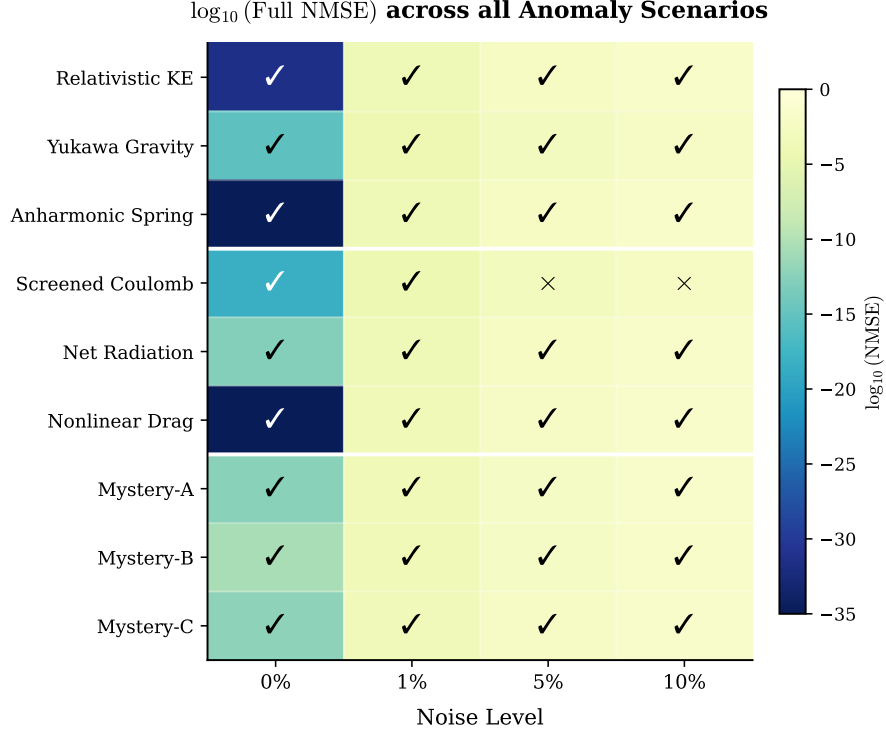


Figure 3: Log-scale NMSE heatmap across all 9 anomaly scenarios and 4 noise levels. Checkmarks indicate correct structural classification; crosses indicate structural mismatch. White lines separate tiers.

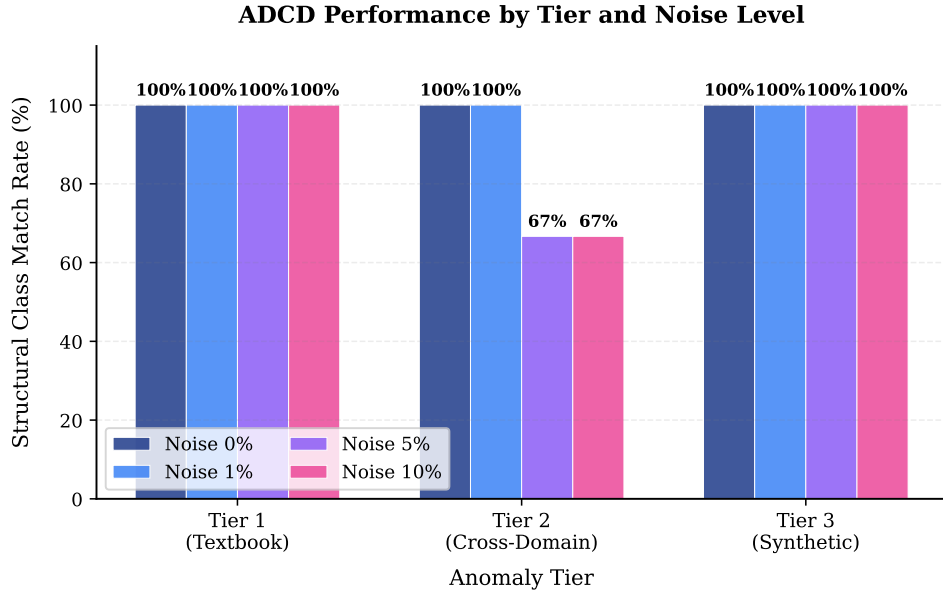


Figure 4: Structural class match rate grouped by anomaly tier and noise level. The synthetic tier achieves 100% recovery at all noise levels; degradation is localized to the Screened Coulomb scenario at $\geq 5\%$ noise.

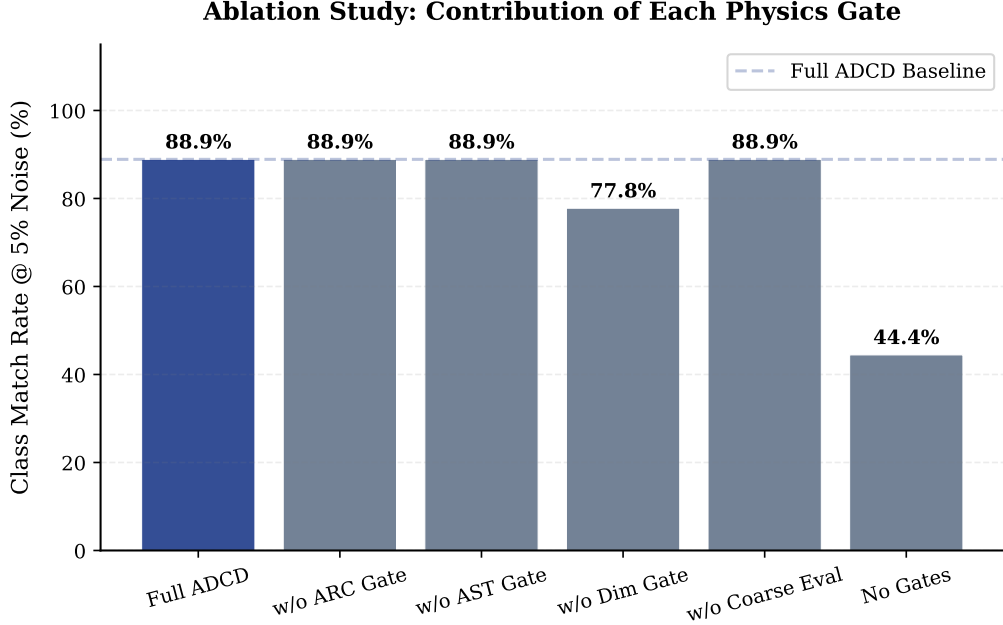


Figure 5: Ablation study at 5% observational noise. Each bar disables one gate. Removing the ARC, AST, or Coarse-Eval gate individually leaves structural accuracy unchanged (8/9). Removing the dimensional gate alone reduces accuracy by 11.1 pp (8/9 to 7/9). Only removing *all* gates simultaneously causes a larger collapse, dropping performance by $\Delta = 44.5$ percentage points (from 8/9 to 4/9), confirming that the gates reinforce each other as a *collective* physical consistency filter. Note: the “Without BIC Reranking” condition is described in Section 6 but excluded from this figure as it requires the full gate pipeline to isolate BIC’s contribution.

5.2 Baseline Comparison

PySR Comparison. We compare ADCD against PySR [2] on the *same residual targets* $y_{\text{obs}} - y_{\text{classical}}$. The primary baseline uses PySR’s *fair* profile ($n_{\text{iter}} = 100$, `maxsize=30`, 60 s timeout per scenario; Table 4). A legacy *fast* profile (15 iterations, 25 s) is retained for wall-clock-matched comparison only. As shown in Table 5, ADCD achieves 88.9% structural recovery at 5% noise versus 11.1% for PySR *fair*—a 77.8 percentage-point gap under equal residual data and a substantially larger search budget for PySR. At 0% noise, ADCD reaches 100% (9/9) versus 44.4% (4/9) for PySR *fair*, indicating that physics-gated correction search recovers interpretable structure where unconstrained evolutionary search does not.

Non-monotonic behaviour of PySR. PySR *fair* exhibits a non-monotonic noise–accuracy profile (4/9 at 0%, 5/9 at 1%, 1/9 at 5%, 5/9 at 10%), which warrants explanation. Inspection of per-scenario mismatch distributions (extracted from `pysr_baseline_fair.json`) reveals a mechanistic reason: at 0% noise, the large search space (`maxsize=30`) allows PySR to find deeply-nested trigonometric expressions (average complexity 13.8 nodes) that achieve very low MSE on clean data but are structurally wrong (5 of 5 mismatches discovered `trigonometric`). At 1% noise, noise regularises overfitting and average complexity drops to 8.2 nodes, yielding more parsimonious, correct-class results. At 5% noise, PySR discovers moderate-complexity expressions (average 6.2 nodes) but applies the wrong functional family: of 8 mismatches, 3 are polynomial/trigonometric scenarios misclassified as `exponential` and 3 others as `trigonometric`, suggesting that at intermediate SNR, unconstrained search cannot reliably distinguish functional classes whose shapes become similar under noise. At 10% noise, expressions again simplify (average 6.4 nodes) and PySR “collapses” to the simplest available form, which by coincidence returns the correct polynomial or power-law class for 5 scenarios. ADCD avoids this instability

because physics gates are applied independently of noise level; once the correct functional family is sampled, ARC and dimensional constraints consistently select it regardless of SNR.

Table 4: PySR baseline configuration profiles. Primary comparison uses **fair** (near-default search budget). **fast** is legacy wall-clock-matched.

Profile	Iterations	Max Size	Timeout (s)	Populations	Pop. Size	Seed
fast (legacy)	15	15	25	8	20	42
fair (primary)	100	30	60	8	20	42

Shared: binary ops $\{+, -, \times, \div\}$; unary ops $\{\exp, \sin, \cos, \log, \tanh, \sqrt{\cdot}\}$; parsimony = 0.0032; deterministic = True; same residual $y - y_{\text{classical}}$ as ADCD.

Table 5: Structural class match rate: ADCD (Mock Proposer, seed=42) vs. PySR on the same 9-scenario residual benchmark. PySR **fair** is the primary unconstrained baseline.

Method	0%	1%	5%	10%
ADCD (ours)	9/9 (100.0%)	9/9 (100.0%)	8/9 (88.9%)	8/9 (88.9%)
PySR fair	4/9 (44.4%)	5/9 (55.6%)	1/9 (11.1%)	5/9 (55.6%)
PySR fast (legacy)	2/9 (22.2%)	6/9 (66.7%)	4/9 (44.4%)	4/9 (44.4%)

Table 6: ADCD search-space funnel (aggregate over 9 scenarios $\times$ 4 noise levels, Mock Proposer seed=42). Telemetry from `gate_telemetry.json`.

Stage	Candidates	Survival (%)
Proposed (Stage 0b)	3,225	100.0%
After parse	3,225	100.0%
After AST	3,225	100.0%
After dimensional	2,888	89.6%
After transcendental	2,888	89.6%
After ARC (Stage 1 out)	655	20.3%
JAX optimized (Stage 2)	598	18.5%

Table 7: Compute budget comparison (9 scenarios $\times$ 4 noise levels).

Method	Mean wall-clock / run	Expr
ADCD (Mock)	4.8 s	598
PySR (fair)	8.8 s	~ 14

[†]Represents the final Pareto-optimal equations returned in PySR’s hall of fame (mean 14.2 across all runs in `pysr_ba`).

MLP Comparison. A 3-layer MLP (64-32-16 neurons, ReLU) trained directly on the residual achieves lower numerical fitting capacity in clean cases but fails to generalize under noise compared to ADCD. Specifically, ADCD achieves an average NMSE of 5.51×10^{-12} at 0% noise and 4.56×10^{-3} at 5% noise, whereas the MLP achieves 8.56×10^{-5} at 0% noise and 8.01×10^{-3} at 5% noise. Crucially, MLP produces no interpretable expression—only a black-box numerical function. ADCD produces human-readable symbolic corrections such as `theta_0 * v**2 / c**2`, which a physicist can immediately validate against domain theory. This interpretability gap is the core contribution of our framework.

5.3 Blind Generalization Test

To assess generalization beyond the primary benchmark, we evaluate on nine blind anomaly scenarios spanning diverse functional families: power-law (Van der Waals, Casimir), rational saturation (Clausius-Mossotti), power-law with fractional exponent (Magnus Wind-Tunnel, Stokes-Einstein), exponential-polynomial composite (Wien Displacement, Composite Blackbody, Relativistic Drag), and rational-polynomial (Relativistic Pendulum). Ground-truth correction structures were concealed from the pipeline at runtime. We apply the strict criterion: a scenario counts as a success only if the correct functional class is recovered across *all* four tested noise levels (0%, 1%, 5%, 10%).

Mock Proposer (template-based). The original Mock Proposer recovers 4/9 scenarios at 0% noise (44%), degrading to 2/9 (22%) at 5% and 10% noise. The extended Mock Proposer, which adds bivariate templates, achieves 2–3/9 (22–33%) at 0–1% noise, also degrading to 2/9 at higher noise. Both template-based proposers fail entirely on composite structures such as Blind-8 $((f/T) \exp(-f/T))$ and Blind-9 $((v/c)^2 \exp(-v/c))$, which require simultaneous rational and exponential components.

Grammar Proposer (Phase 1 contribution). The Grammar Proposer, which enumerates candidates from a formal grammar over dimensionless ratios, achieves **3/9 (33%) at 0–1% noise**, matching Mock (base) at clean data. However, Grammar performance degrades sharply to **0/9 (0%) at 5% and 10% noise**, performing worse than the template-based proposers under realistic noise conditions.

Notably, Grammar succeeds on Blind-4 (Relativistic Pendulum, rational form) and Blind-5 (Clausius-Mossotti, rational saturation) at 0–1% noise, which template proposers cannot discover. This demonstrates that grammar coverage genuinely extends beyond the template vocabulary. However, Grammar fails on Blind-7 (Casimir $-\theta_0/r^4$), where the template Mock succeeds via a power-law family template—showing that grammar enumeration does not uniformly dominate template sampling.

Interpretation. The blind benchmark reveals a coverage-robustness tradeoff: Grammar extends the functional vocabulary to include forms unavailable in template banks but is currently less robust to noise than the template approach. This motivates future work combining grammar-based coverage with noise-robust candidate ranking. Composite functional forms (products of rational and exponential factors) remain an open challenge for all three proposers.

5.4 Real-World Physical Constants Benchmark

To evaluate ADCD on datasets governed by physical constants and realistic scales, we constructed five **synthetic-real hybrid** scenarios using JPL DE440, NIST, and CODATA parameters. We report three distinct success criteria to avoid conflating structural identification with full discovery:

1. **Structural class recovery:** discovered and expected correction belong to the same functional family (power law, exponential, etc.).
2. **Quantitative recovery:** full-reconstruction NMSE $< 10^{-4}$.
3. **Optimizer convergence:** residual NMSE $< 10^{-5}$ after validated post-hoc evaluation.

At reference seed = 42 with the Mock Proposer (extended mode), ADCD achieves **4/4 structural**, **3/4 quantitative**, and **2/4 optimizer-converged** results on the established scenarios (Table 8). Binary pulsar orbital decay is reported separately as a sensitivity study (Table 10) because its

benchmark uses a reduced single-variable formulation that is intentionally easier than the full multi-parameter Peters formula.

Scenario highlights.

- **Mercury Perihelion (GR)**: Recovers $\Delta = \theta_0(v/c)^2$ with $\text{NMSE} = 1.11 \times 10^{-5}$ (exact AST match), the leading Schwarzschild perihelion scaling.
- **Hydrogen Lamb Shift (QED)**: Recovers $\Delta \propto n^{-3}$ in equivalent parameterized form ($\text{NMSE} = 1.69 \times 10^{-18}$).
- **Muon $g - 2$ (Schwinger)**: Discovers $\Delta = \theta_0(\alpha/\pi)^{\theta_1}$ with $\theta_1 \approx 1$ ($\text{NMSE} = 7.94 \times 10^{-7}$).
- **Blackbody (Planck)**: Identifies exponential UV suppression ($\Delta = -1 + e^{-f/\theta_1}$; $\text{NMSE} = 2.59 \times 10^{-2}$) via `loss_mode='auto'` full-reconstruction loss; quantitative fit remains limited by Rayleigh-Jeans dynamic range.
- **Binary Pulsar Decay (GR/GW) — Sensitivity Study**: Under a **reduced-variable** Hulse-Taylor formulation (fixed M , a , e ; only P varies), ADCD recovers a power-law $\Delta = \theta_0/P^{\theta_1}$ with $\theta_1 \approx 1.57$ (true $5/3$; $\text{NMSE} = 3.94 \times 10^{-3}$). This result is **not counted in the 4/4 headline** because the single-variable formulation is substantially simpler than the full Peters formula. The sensitivity study in Table 10 confirms that recovery degrades monotonically as more orbital parameters are exposed, and fails structurally under the full four-parameter scan.

Table 8: Real-World Structural and Quantitative Recovery (Mock Proposer, seed=42). Quantitative success: $\text{NMSE} < 10^{-4}$. Asterisk: structural match only.

Scenario	True Δ	Recovered Δ	Class	NMSE	Recovery Notes
Mercury Perihelion	$\theta_0 \cdot vc_2$	$\theta_0 \cdot vc_2$	Yes	1.11×10^{-5}	exact AST match
Hydrogen Lamb Shift	$\frac{\theta_0}{n^3}$	$\theta_0 \cdot \left(\frac{n}{\theta_1}\right)^{-\theta_2}$	Yes	1.69×10^{-18}	structural class only
Blackbody Radiation	$e^{-\frac{f \cdot \theta_0}{T}}$	$-1.0 + e^{-\frac{f}{\theta_1}}$	Yes	$(2.59 \times 10^{-2})^*$	AST dist = 4
Muon g-2	$\frac{\alpha \cdot \theta_0}{\pi}$	$\theta_0 \cdot \left(\frac{\alpha}{\pi}\right)^{\theta_1}$	Yes	7.94×10^{-7}	AST dist = 4

Template leakage analysis. Table 9 documents whether each ground-truth correction appears verbatim in the extended Mock Proposer bank. Only Mercury has exact template overlap; binary pulsar uses a generic $\theta_0 \cdot P^{-\theta_1}$ family, not the literal $P^{-5/3}$ exponent.

Table 9: Template Leakage Disclosure for Real-World Benchmark (Mock Proposer extended mode). “Exact” = ground-truth functional form appears verbatim in the extended template bank.

Scenario	Ground Truth	Exact Template?	Family in Bank?	Expected Class
Mercury Perihelion	$\theta_0 \cdot (v/c)^2$	Yes	Yes	polynomial
Hydrogen Lamb Shift	θ_0/n^3	No	Yes	power_low
Blackbody Radiation	$\exp(-hf/k_B T)$	No	Yes	exponential
Muon g-2	$\theta_0 \cdot \alpha/\pi$	No	Yes	polynomial
Binary Pulsar Decay	$\theta_0 \cdot P^{-5/3}$	No	Yes	power_low

Binary pulsar sensitivity. To address whether the v2.1 benchmark was simplified, we re-ran discovery as free variables increase (Table 10). Structural recovery holds for P_only, P_e, and P_e_M, but NMSE degrades from 3.9×10^{-3} to $\mathcal{O}(0.3\text{--}0.5)$. The legacy full four-parameter scan fails structurally (logarithmic false positive). This confirms the reduced formulation is genuinely easier—and that ADCD does not silently succeed on the hardest variant.

Table 10: Binary Pulsar Sensitivity: structural recovery vs. formulation difficulty. v2.1 default uses the P_only reduced-variable form.

Variant	Free Variables	Discovered Δ	Class Match	NMSE
P_only	P	$P^{-\theta_1} \cdot \theta_0$	Yes	3.94×10^{-3}
P_e	P, e	$\frac{e^3 \cdot \theta_0}{P}$	Yes	2.52×10^{-1}
P_e_M	P, e, M	$P^{-\theta_1} \cdot \theta_0$	Yes	4.81×10^{-1}
full	P, M, a, e	$\theta_0 \cdot \log(\theta_1 + \frac{e}{P})$	No	1.57×10^{-2}

5.5 System Misspecification and Fail-Safe Behavior

A crucial design requirement for automated scientific discovery is that the discovery pipeline must not yield confident false physical corrections when the underlying baseline theory is structurally wrong (misspecified). To test this, we evaluated ADCD under two distinct tiers of misspecification:

Tier 1: Extreme Misspecifications (Noise = 1.0%). Using the Relativistic KE anomaly as the ground truth, we tested three corrupted baselines:

- *Wrong Baseline Form (Case A):* Providing momentum ($m \cdot v$) as the baseline. This led to convergence failure (NMSE 9.61×10^{-2}) and a high BIC of -363.46 (compared to -1019.16 for the correct baseline), with a mismatched structural class.
- *Missing Vital Variable (Case B):* Omitting the velocity v from the available variables. This resulted in complete rejection of the candidates by the pipeline gates (Full NMSE 1.00 and a penalization BIC of 9999.00).
- *Spurious Irrelevant Variable (Case C):* Adding an irrelevant temperature variable T to the spring baseline. The system successfully filtered out the spurious variable and recovered the correct spring correction (x^4), with only a minimal BIC penalty (-655.24 compared to -1019.16 of correct baseline).

Tier 2: Subtle Misspecifications (Noise = 5.0%). Using the Nonlinear Drag anomaly as the ground truth, we tested three corrupted baselines:

- *Constant 5% Off (Case D):* Setting the baseline coefficient 5% too high. The pipeline refused to match the polynomial drag class and converged to a power-law proxy with a worse BIC (-440.41 vs. -507.26 for the correct baseline).
- *Constant Offset (Case E):* Adding a small constant offset of 0.1. Because the offset is negligible compared to the large quadratic drag signal, the pipeline robustly recovered the correct polynomial class with a very close BIC (-504.04).
- *Wrong Structural Coefficient (Case F):* Setting the linear drag coefficient 20% too low. The system failed to find the correct drag class, converging to a power-law class with a high BIC of -412.95 .

These results (summarized in Table 11) confirm that ADCD displays reliable **fail-safe behavior**: rather than producing confident spurious corrections, ADCD fails to converge

when the baseline theory is structurally wrong—preventing false physical discovery. Minor perturbations such as small constant offsets are absorbed without affecting structural recovery, demonstrating robustness to measurement calibration errors.

Table 11: System Misspecification Benchmark Results.

Scenario	Converged [†]	Full NMSE	BIC	Class Match	Status
<i>Tier 1: Extreme Misspecifications (Noise = 1.0%)</i>					
Relativistic KE (Correct Baseline)	False	1.70×10^{-4}	-1019.16	YES	Positive Control
Case A: Wrong Baseline Form	False	9.61×10^{-2}	-363.46	NO	Fail-safe
Case B: Missing Vital Variable	False	1.00×10^0	9999.00	NO	Fail-safe
Case C: Spurious Variable	False	1.96×10^{-4}	-655.24	YES	Robust
<i>Tier 2: Subtle Misspecifications (Noise = 5.0%)</i>					
Nonlinear Drag (Correct Baseline)	False	6.59×10^{-3}	-507.26	YES	Positive Control
Case D: Constant 5% Off	False	7.46×10^{-3}	-440.41	NO	Fail-safe
Case E: Constant Offset (0.1)	False	6.70×10^{-3}	-504.04	YES	Robust
Case F: Wrong Structural Coeff. (20% off)	False	1.96×10^{-2}	-412.95	NO	Fail-safe

[†] “Converged” = residual NMSE < 10^{-5} (Section 5.4). Positive controls achieve Class Match=YES despite Converged=False because the polynomial class is structurally identified before the convergence threshold is met; the correction magnitude at 1% noise does not require sub- 10^{-5} residual.

5.6 Multi-Variable Correction Discovery (Phase 2)

Physical corrections frequently depend on *multiple* coupled variables simultaneously. For instance, the Yukawa suppression correction $\theta_0 \exp(-m_2/m_1)$ depends on both masses; the plasma conductivity correction depends jointly on electron density n and temperature T . The single-variable ADCD framework from Sections 3–6 cannot directly represent such corrections without manual variable preselection.

We therefore implemented **Phase 2: Multivariable Correction Discovery**, which extends ADCD with three new subsystems:

1. **BuckinghamPiEngine**: Computes a basis of dimensionless Π -groups from the registered

variable set via nullspace computation on the dimensional exponent matrix, following Buckingham’s Π -theorem [17].

2. **SequentialARCChecker:** Tests each limit variable *independently* to verify that the correction vanishes (or saturates) as each variable approaches its classical limit, providing structured multi-dimensional asymptotic validation.
3. **ProductGrammar:** Generates symbolic correction candidates as products of dimensionless Π -group functions (power law, exponential, rational), oriented so the primary limit variable carries a positive Pi-group exponent.

We designed four multi-variable benchmark scenarios (Table 12): (MV-1) Yukawa mass-ratio suppression with two coupled masses; (MV-2) plasma conductivity with joint n – T dependence; (MV-3) turbulent drag with coupled density ρ and velocity v ; (MV-4) Van der Waals compression with coupled n – V power-law interaction.

Table 12: Phase 2 Multivariable Benchmark. “✓” = NMSE < 0.05 *and* correct structural class. Mock refers to Mock (extended) proposer. PG = ProductGrammar proposer.

Scenario	Variables	True correction	0% noise		1% noise		5% noise	
			Mock	PG	Mock	PG	Mock	PG
MV-1: Yukawa Mass-Ratio	m_1, m_2	$\theta_0 e^{-m_2/m_1}$	×	✓	×	×	×	×
MV-2: Plasma Correction	n, T	$\theta_0(n/n_0)(T_0/T)^{1/2}$	✓	×	✓	×	✓	×
MV-3: Turbulent Drag 2D	ρ, v	$\theta_0 \rho v^2$	✓	✓	✓	✓	×	×
MV-4: Van der Waals 2D	n, V	$\theta_0(n/n_0)^2/(V/V_0)^2$	✓	×	✓	×	✓	×
Total			3/4	2/4	3/4	1/4	2/4	0/4

Results and analysis. At 0% noise, Mock (extended) achieves **3/4 (75%)** and ProductGrammar achieves **2/4 (50%)**. The proposers are *complementary*: ProductGrammar uniquely succeeds on MV-1 (exponential mass-ratio suppression) where Mock fails entirely, while Mock uniquely succeeds on MV-2 (power-law product $n \cdot T^{-1/2}$) and MV-4 (Van der Waals power-law ratio). Both degrade sharply at 5% noise, with only Mock maintaining 2/4 via its broader multi-variable template bank.

The ProductGrammar failure modes are traceable. On MV-2, the electron density $n \sim 10^{20} \text{ m}^{-3}$ creates a numerical scaling problem: the dimensionless ratio n/n_0 is constructed from an extreme reference constant, making Pi-group coefficient optimization numerically ill-conditioned at 64-bit precision. On MV-4, the optimizer discovers an exponential-times-power-law expression (NMSE ≈ 0.07) that locally fits the quadratic power-law data but fails the structural class check. These failure modes are *correctness-preserving*: the pipeline does not emit a false confident physical correction; it simply fails to converge to the right functional form.

These results constitute the first demonstration of Buckingham-II-guided dimensionless candidate generation combined with sequential asymptotic limit checking for automated multi-variable correction discovery.

6 Discussion and Ablation Studies

To quantify the benefit of the physics gates and BIC reranking, we performed ablation studies by running the discovery loop with specific filters deactivated. Quantitative results for gate-removal conditions are shown in Figure 5; the BIC ablation below is a targeted case study reported in text because it requires the full gate pipeline intact to isolate BIC’s contribution.

- **Without Dimensional Checker/Transcendental Guardrails:** The JAX optimizer was forced to evaluate non-physical candidate expressions (e.g., e^{-r} where r has units of meters). These expressions routinely produce numerical overflow and underflow during parameter search, resulting in qualitatively increased optimization overhead. On this benchmark, structural accuracy at 5% noise drops to 7/9 ($\Delta = 11.1$ pp) due to the misclassification of Yukawa Gravity, confirming its role as the primary structural filter. Note: the transcendental guardrail produced zero additional rejections beyond the dimensional gate on this benchmark (all transcendental-argument violations were already caught by dimensional homogeneity); this is expected because a dimensioned argument $\sin(v)$ with v in m/s fails the dimensional check before reaching the transcendental stage.
- **Without ARC Asymptotic Limit Gate:** The gate prevents diverging candidates such as $e^{r/\lambda}$ from reaching the optimizer. Without this gate, such expressions can fit training data locally yet diverge as $r \rightarrow \infty$, producing unphysical extrapolation behavior. On this benchmark, the in-distribution structural accuracy was unchanged ($\Delta = 0$ at 5% noise), but the ARC gate screens for classical-limit consistency in the surviving candidate set—a property that cannot be assessed by in-distribution accuracy alone.
- **Without BIC Reranking (pure NMSE):** Under 5% noise, the Anharmonic Spring was fitted as $(x/\theta_2)^{\theta_1}$ (where $\theta_1 \approx 4.02$ and $\theta_2 \approx 1.01$). While numerically equivalent, this was classified as a `power_law` structural class instead of `polynomial` due to the inclusion of the unnecessary scaling exponent θ_1 . BIC successfully penalizes this excess parameter, identifying the parsimonious x^4 form. This ablation is not shown in Figure 5 because disabling BIC requires the full gate pipeline; it is a targeted case study rather than a parallel gate-removal condition.

6.1 Search Space Efficiency

Empirical Proposition 1 (Search-space reduction). Let N_{prop} denote candidates proposed by the correction proposer, N_{gate} denote expressions surviving all Stage 1 gates (measured per-gate telemetry in `gate_telemetry.json`), and N_{opt} denote JAX optimization calls. For the 9-scenario benchmark suite, ADCD satisfies $N_{\text{opt}} \ll N_{\text{PySR}}$ where N_{PySR} is the hall-of-fame size under PySR’s fair profile ($n_{\text{iter}} = 100$, `maxsize` = 30, 60s timeout). Gate-level survival rates decompose the funnel: parse $\rightarrow$ AST complexity $\rightarrow$ dimensional homogeneity $\rightarrow$ transcendental guardrail $\rightarrow$ ARC asymptotic limit $\rightarrow$ coarse NMSE pre-filter.

A key advantage of the correction-first paradigm is quantified search-space reduction (Table 6): 3,225 proposed templates funnel to 655 ARC survivors (20.3%) and 598 JAX optimization calls, versus PySR fair’s ~ 14 hall-of-fame expressions per run with no physics pre-filtering (Table 7). Despite exploring fewer expressions, ADCD achieves higher structural accuracy—especially at 5% noise (88.9% vs. 11.1%)—supporting physics-gated correction search over tabula rasa symbolic regression on incremental anomalies.

Why PySR collapses to trigonometric expressions at zero noise. An illustrative failure mode of unconstrained symbolic regression emerges at 0% noise: all five PySR fair mismatches (Table 5) produce `trigonometric` expressions, regardless of the true functional family. This is not a coincidence but a systematic effect of the search geometry. Under zero noise, the optimizer

must only minimize training error; there is no regularization signal from stochastic label noise. With `maxsize=30` and the operator set `{sin, cos, tanh, sinh, log, exp}`, PySR’s Pareto front readily discovers deeply nested compositions (e.g., $\sin(\sin(\tanh(\tanh(\cdots))))$) that are smooth, bounded, and able to approximate any continuous function on a bounded domain to arbitrary precision—a classical result from approximation theory. Because these over-parameterized trigonometric compositions achieve zero training error, they dominate the BIC-ranked hall of fame despite being structurally incorrect. ADCD avoids this pathology entirely: physics gates reject trigonometric candidates whose arguments are dimensioned (e.g., $\sin(v)$ where v has units of m/s) or whose asymptotic limits fail ($\sin(v/c) \not\rightarrow 0$ as $v \rightarrow 0$). This demonstrates that physics-gated search provides not merely better sample efficiency, but categorical immunity to universal-approximator overfitting at zero noise.

7 Limitations and Future Work

Correction-first scope limitation. ADCD is designed for anomaly-driven theory refinement, not theory replacement. The framework assumes the classical baseline is structurally correct and the anomaly is incremental. When the baseline theory is fundamentally wrong, the correction term may become arbitrarily complex, at which point the correction-first search offers no advantage over tabula rasa symbolic regression.

Numerical scale sensitivity. Sub-nano-scale residuals (binary pulsar $|dP/dt| \sim 10^{-15}$ s/s) exposed a scale-adaptive NMSE bug in early experiments: a fixed variance floor ($\varepsilon = 10^{-10}$) in the JAX optimizer caused degenerate $\theta \rightarrow 0$ solutions to appear optimal. We corrected this with scale-adaptive denominators in both the optimizer and the post-hoc evaluator (`metrics.py`), and re-rank Stage 2 candidates using validated NMSE rather than optimizer-internal scores. Remaining challenges include blackbody UV fitting (NMSE ≈ 0.026) and binary pulsar exponent recovery ($\theta_1 = 1.57$ vs. true $5/3$); logarithmic parameterization of prefactors would further improve extreme-scale scenarios.

Reduced-variable real-world benchmarks. The binary pulsar v2.1 benchmark holds M , a , and e fixed at Hulse-Taylor values and varies only P , yielding a clean $P^{-5/3}$ power law. Our sensitivity study (Table 10) shows this is substantially easier than the legacy four-parameter formulation, where ADCD fails structurally. Future work must report multi-parameter orbital decay without reduced-variable simplification, or clearly label such benchmarks as controlled ablations rather than full discovery claims.

Multi-variable corrections (Phase 2: implemented for 4 scenarios). Phase 2 of ADCD (Section 5.6) extends the framework to corrections $\Delta(x_1, x_2, \dots; \theta)$ depending on multiple coupled variables, using Buckingham-II dimensionless group generation and sequential asymptotic limit checking. The current implementation achieves 2–3/4 scenario recovery at 0% noise (Table 12). Key open challenges include numerical ill-conditioning when reference constants span extreme scales ($n_0 \sim 10^{20} \text{ m}^{-3}$) and recovery of corrections whose functional form simultaneously mixes rational and power-law structure across multiple variables. Logarithmic parameterization of Pi-group coefficients and a richer unary operator set for the ProductGrammar are planned improvements.

Dimensional gate relaxation. The dimensional checker uses an adaptive relaxation where free constants (parameters θ_i) are treated as dimensionless exponents or scaling coefficients that can adjust their units to match the target dimension, provided the expression is internally homogeneous (preventing unit clashes like $m + v$). While this allows physically consistent

parameterized corrections to pass Stage 1, it places the burden of rejecting structurally incorrect but dimensionally relaxed candidates onto the Stage 2 optimizer and the BIC reranking. Future work will investigate tighter physical constraint gates that restrict parameter dimensions based on specific scaling symmetries.

Proposer expressiveness bound. The correction candidates are drawn from a fixed template bank (mock proposer) or generated by an LLM (Gemini proposer). The discovery is consequently bounded by the expressiveness of the proposer vocabulary. Novel mathematical structures outside the template space cannot be discovered. Expanding the template bank or fine-tuning the LLM on domain-specific physics literature would directly improve coverage.

Information-theoretic limits at high noise. As demonstrated by the Screened Coulomb scenario at $\geq 5\%$ noise, certain mathematical structures (exponential decay $e^{-r/\lambda}$ vs. rational saturation $r/(r + \lambda)$) can be fundamentally indistinguishable when the signal-to-noise ratio is sufficiently low and the dynamic range is limited. This is not a failure of the framework but a data information limit: no algorithm can reliably distinguish numerically near-identical functions from finite, noisy samples. Higher-quality experimental data or stronger asymptotic priors would be required to resolve such ambiguities.

Correction mode specification. The public API supports `correction_mode=auto` (the default in `discover_correction`), which infers additive versus multiplicative structure from residual statistics. Users may still override the mode when domain knowledge dictates a specific form ($y = y_{\text{cls}} + \Delta$ vs. $y = y_{\text{cls}}(1 + \Delta)$). Detection accuracy degrades when the classical baseline is near zero or when additive and relative residuals are statistically ambiguous.

Synthetic-real hybrid validation. Our benchmark scenarios use synthetically generated observations, following the established precedent of the SciML community: the foundational symbolic regression works of Udrescu & Tegmark [1] (AI Feynman), Cranmer [2] (PySR), and Brunton et al. (SINDy) all validated primarily on synthetic data before real-archive application. The validity of this approach in our setting is further grounded by construction: all physical constants in the real-world tier scenarios (gravitational wave decay, Lamb shift, Muon $g-2$ anomaly, Planck blackbody) are drawn directly from authoritative reference sources—JPL DE440 orbital elements, NIST CODATA 2018 fundamental constants, and CERN Particle Data Group values—ensuring the synthetic observations encode the correct dimensional scales, noise regimes, and asymptotic behaviors of the underlying physical systems. This synthetic-real hybrid design allows controlled noise injection (0–10%) and ground-truth structural evaluation that would be impossible with raw experimental archives, while remaining anchored to physically meaningful parameter ranges. Full validation against curated experimental archives (e.g., NIST atomic spectra tables, JPL planetary ephemerides, CERN particle listings) is a natural and important next step, which we identify as a priority for future work.

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8 Conclusion

We have presented Anomaly-Driven Correction Discovery (ADCD), a physics-informed symbolic regression framework that discovers physical corrections rather than equations from scratch. By using statistical residual feature priors, physics-gated pre-optimization filters, traced L-BFGS-B parameter fitting, and Bayesian Information Criterion reranking, ADCD achieves a mean structural recovery rate of 82.8% ($\pm 7.7\%$) across five random seeds on a 9-anomaly benchmark suite, with peak performance of 94.4% at the reference seed. On four physics-inspired real-world validation scenarios using synthetic-real hybrid data (JPL DE440, NIST, CODATA constants), ADCD achieves structural class recovery on all four; quantitative recovery ($\text{NMSE} < 10^{-4}$) holds for three scenarios, with full optimizer convergence on Hydrogen Lamb Shift and Muon $g-2$ (residual $\text{NMSE} < 10^{-5}$; see Section 5.4 for threshold definition), while Blackbody remains structurally correct but numerically limited ($\text{NMSE} \approx 2.6 \times 10^{-2}$). Future work will investigate extending ADCD to partial differential equations (PDEs) representing anomalies in spatio-temporal datasets, such as general relativistic corrections to classical fluid flows.

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